

Automatic Recognition of Quranic Tajweed Rules Across Riwayat: A Corpus and Machine-Learning Study of Idgham and Qalqalah

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Abstract. Automatic assessment of Quranic recitation requires systems that determine not only *what* was recited but *how* it was pronounced. We present a riwayat-aware framework for the automatic recognition of Tajweed rules from speech. A corpus of 7,998 verse-level recordings (11.17 h) was built from MP3Quran recitations of Juz' 'Amma by fifteen reciters covering the riwayat of Hafs 'an 'Asim, Warsh 'an Nafi', and Qalun 'an Nafi'; verse segments were obtained by forced alignment with manual reconciliation, and occurrences of Idgham and Qalqalah were localised through rule-based candidate extraction, manual acoustic annotation and expert verification (705 unique textual occurrences). Recitation segments are described by eight interpretable acoustic descriptors and classified into five Tajweed categories with SVM, random forest, k -nearest neighbours, XGBoost and a multilayer perceptron. The proposed representation outperforms an MFCC baseline with deltas for every classifier, reaching a Macro-F1 of 0.6935 with XGBoost and gains of up to +0.1348. Ablation identifies duration and a depth metric as the dominant cues, SMOTE raises Idgham bila-Ghunnah recall from 0.0678 to 0.4915, and cross-riwayat transfer yields Macro-F1 values of 0.6584–0.6718.

Keywords: Quranic recitation , Tajweed recognition , Idgham , Qalqalah , acoustic feature extraction , machine learning , riwayat-aware classification

1 Introduction

Qur'anic recitation is a highly structured form of oral Arabic whose pronunciation is governed by the principles of Tajweed and normally acquired through repeated practice with expert feedback. Conventional assessment therefore depends on expert listening, making frequent, scalable and objective evaluation difficult outside formal instruction. Recitation is further transmitted through recognised *riwayat*—narrations sharing the same Qur'anic text but differing in phonetic, articulatory and recitational detail—so a computational system must treat legitimate riwayat variation as distinct from pronunciation error. This study considers the “Big Three” riwayat: Hafs 'an 'Asim, Warsh 'an Nafi', and Qalun 'an Nafi'.

Automatic assessment of Qur'anic recitation differs fundamentally from conventional speech recognition, which determines what was recited rather than how particular phonetic sequences were realised; reciter identification models speaker characteristics, and generic pronunciation assessment provides no explicit rule-level

representations. The central challenge is thus to detect specific Tajweed phenomena from the acoustic signal despite speaker variability, riwayat-related differences and scarce annotated resources—a challenge for which appropriately annotated speech data is a prerequisite that existing Qur'anic corpora do not meet (Sect. 2).

Idgham and Qalqalah offer complementary test cases. Idgham is an assimilation between adjacent consonants that modifies the temporal and spectral structure of the affected boundary, optionally with nasalisation (ghunnah); Qalqalah is an echo-like release of the consonants *qaf*, *ta'*, *ba'*, *jim*, *dal* in sukun states. Their contrasting signatures make them suitable for testing whether conventional acoustic descriptors capture rule-specific information while remaining robust across riwayat.

Prior work leaves four aspects insufficiently explored: the dominance of lexical- and speaker-level objectives over explicit Tajweed modelling; the scarcity of studies treating a rule occurrence—rather than an utterance—as the unit of analysis; the absence of rule-level temporal annotations; and the unexamined influence of multiple riwayat on acoustic realisation and detection. This paper therefore treats corpus construction, riwayat identification, rule-level annotation, feature extraction, supervised classification and evaluation as interconnected components of one problem.

The contributions are fourfold. *Resource*: a Qur'anic speech corpus covering the Big Three riwayat, annotated at the Tajweed-rule level. *Method*: a riwayat-aware pipeline connecting annotation with acoustic feature extraction and supervised classification. *Experimental*: a quantitative evaluation of five classifiers on Idgham and Qalqalah across the three riwayat. *Application*: a foundation for computer-assisted recitation systems providing rule-specific, interpretable feedback while respecting legitimate riwayat variation.

2 Related Work

2.1 Qur'anic Speech Resources and Recognition

Existing Qur'anic speech resources differ considerably in scale, speaker diversity, recording conditions, annotation granularity and intended use (Table 1). Verse-level repositories such as EveryAyah [1] link recordings to verses but provide no temporal Tajweed annotation, and the large-scale ASR corpora—the 581,216-recording Quran Recitations corpus [2] and the speaker-disjoint Quran-Ayah-Corpus [3]—prioritise transcription over pronunciation correctness. Diversity-oriented resources (Tarteel [4], Ar-DAD [5], QDAT [6] and the crowdsourced corpus of non-Arabic-speaking reciters [7]) contribute speakers, realistic conditions or expert judgements, but only text- or category-level supervision. Transmission-aware resources are the most recent step: AQQD [8] adds qira'at and phonetic-variation metadata and Riwaya-ID [9] separates Warsh from Hafs with 82% accuracy using wav2vec 2.0 [10] and Whisper [11], yet neither provides temporally localised rule labels. Collectively these resources trade lexical and speaker coverage against rule-level supervision, so corpus size alone cannot satisfy the requirements of automatic Tajweed recognition.

Table 1. Comparison of Qur’anic speech resources. “Rule-level temporal labels” indicates whether the resource localises individual Tajweed phenomena in time.

Resource	Recordings	Speakers	Supervision	Rule-level
EveryAyah [1]	verse-level	professional	verse–audio link	no
Tarteel [4]	$\approx 25,000$ (67.39 h)	$> 1,200$	audio–text	no
Ar-DAD [5]	15,810 + 397	30 + 12	reciter identity	no
QDAT [6]	$> 1,500$	> 150	3 rules, expert	categorical
Crowdsourced [7]	$\approx 7,000$ (1,166 ann.)	1,287	6 categories	categorical
Quran Recitations [2]	581,216 ($\approx 2,882$ h)	44	transcript	no
Quran-Ayah-Corpus [3]	263,263	44	transcript, disjoint splits	no
AQQD [8]	24,183	309	qira’at metadata	partial
Riwaya-ID [9]	> 700 h	–	riwayah label (82%)	no
Ours	7,998 (11.17 h)	15	rule-level, expert-verified	yes

2.2 Automatic Tajweed Recognition

Acoustic Tajweed assessment began with phoneme-duration analysis of Madd and Ghunnah [12] and progressed through MFCC-based statistical and neural classifiers for Qalqalah [13, 14], binary SVM formulations [15], convolutional [16] and recurrent [17, 18] models, and hybrid ASR–classification pipelines [19] (Table 2). Three conclusions recur across these settings: Tajweed phenomena are not equally separable acoustically; accuracy alone is insufficient under class imbalance; and performance obtained under controlled conditions does not imply robust generalisation to unseen speakers, recordings or riwayat. Idgham and Qalqalah nonetheless remain fragmented across datasets, feature representations and evaluation protocols, with no unified framework spanning multiple reciters and transmission traditions.

Table 2. Prior automatic Tajweed recognition systems compared with the present study. Acc.\ = accuracy; val.\ = validation.

Study	Rules	Representation	Model	Data	Reported result
Mohammed et al. [12]	Madd, Ghunnah	duration	threshold	10 reciters	interpretable cues
Qalqalah VQ [13]	Qalqalah	MFCC	vector quant.	–	feasibility
Qalqalah MLP [14]	Qalqalah Kubra	MFCC	MLP	–	feasibility
Smartajweed [15]	4 binary rules	filter bank	RBF-SVM	–	99% val. acc.
Omran et al. [16]	Qalqalah	MFCC	CNN	3,322 / 4 rec.	90.8% val. acc.
Al Harere \	Al Jallad [17]	3 rules	MFCC	LSTM	$>1,500$ (QDAT)
TajweedAI [19]	Qalqalah	ASR + alignment	hybrid	internal/ext.	100% / 57.14%
Ours	2 Idgham + 3 Qalqalah	8 descriptors	5 classifiers	7,998 / 15 rec.	Macro-F1 0.6935

2.3 Research Gap and Positioning

Four gaps define the present work. First, large ASR corpora provide verse-level textual supervision whose objective—transcription—does not encode whether a rule was correctly realised. Second, rule-level temporal annotation identifying *where* a phenomenon occurs in continuous recitation is largely absent, preventing models from learning rule realisations independently of context. Third, rule-specific datasets trade speaker diversity against acoustic control: four professional reciters [16] versus crowdsourced but heterogeneous recordings [6]. Fourth, riwayat is treated as a global classification target [9] rather than as a factor shifting the acoustic realisation of individual rules. The proposed study addresses these gaps by formulating Idgham and Qalqalah recognition as a riwayat-aware, fine-grained acoustic learning problem on a rule-annotated corpus that retains reciter and riwayat identity, bridging large-scale Qur’anic ASR research and rule-specific Tajweed assessment.

3 Methodology

The framework treats automatic Tajweed recognition as a sequence of dependent stages (Fig. 1). Full-surah recordings are aligned and segmented into verse-level clips with explicit reciter and riwayat metadata (Sect. 4); Tajweed occurrences are localised acoustically through a hybrid rule-based, expert-verified annotation protocol (Sect. 5); segments are represented by interpretable acoustic descriptors and analysed for redundancy and imbalance (Sect. 6); and the resulting instances support

a five-class supervised task evaluated with complementary classifiers, ablations, balancing strategies, an MFCC baseline, cross-riwayah transfer and bootstrap uncertainty estimation (Sects. 7–9). Three principles govern the pipeline: the rule occurrence, not the verse, is the unit of analysis; riwayat identity is preserved at every stage so that legitimate variation can be separated from error; and every derived artefact remains linked to the original signal and to the canonical text.

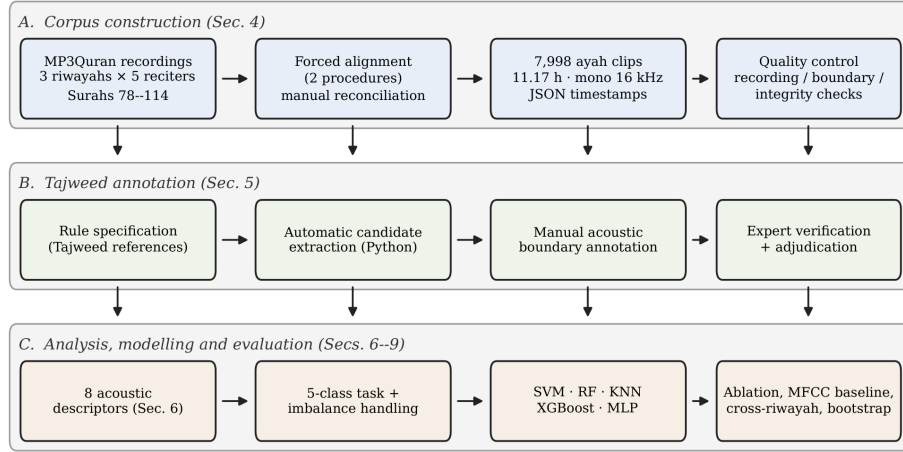


Fig. 1. Overview of the riwayat-aware Tajweed recognition framework: corpus construction with forced alignment and quality control (Sect. 4), hybrid Tajweed annotation (Sect. 5), and acoustic analysis, modelling and evaluation (Sects. 6–9).

4 Corpus Construction

4.1 Source Material and Riwayah Coverage

Audio was collected from the publicly available MP3Quran repository of complete Qur'anic recitations [20]. Rather than restricting the corpus to one tradition, three riwayat were included—Hafs 'an 'Asim, Warsh 'an Nafi', and Qalun 'an Nafi'—each represented by five reciters, giving fifteen reciters in total. Reciter selection prioritised inter-speaker variability (vocal characteristics, articulation, rate, prosody) within a well-defined tradition, because a model trained on a single reciter may learn speaker-specific rather than rule-specific characteristics. The scope is Juz' 'Amma (surahs 78–114), i.e. 37 surahs and 564 ayat per reciter, targeting 555 full-surah recordings; the canonical content corresponds to 8,460 reciter-ayah instances, of which approximately 14.14 h of full-surah audio were collected.

4.2 Segmentation, Alignment and Quality Control

Complete surah recordings were processed with two automatic forced-alignment procedures whose timestamps and recognised text were retained separately and reconciled manually into a consolidated segmentation file per surah, from which ayah-level clips were extracted; this prevents alignment or recognition errors from propagating unchecked into the training data. Quality control operated at three levels: recording level (availability, readability, correspondence with the intended surah and reciter); segmentation level (cross-method comparison and review of boundaries); and data-integrity level (expected surah and ayah counts checked against the canonical structure, with missing recordings identified explicitly rather than treated as complete). Alignment confidence scores are stored to flag potentially unreliable segments.

4.3 Corpus Organisation and Statistics

The corpus separates original recordings from derived segments and metadata: per-riwayah directories contain, for each reciter, an `audio/` folder of full-surah recordings, an `ayat/` folder of verse segments, and JSON timestamp files linking every segment to its source position. After segmentation, alignment and validation, 7,998 verse-level recordings were retained—94.5% of the theoretical maximum—comprising 11.17 h of aligned audio distributed evenly across riwayat (Table 3); for modelling, audio is decoded to mono at 16 kHz. The longer mean verse duration of Warsh and Qalun relative to Hafs is consistent with differences in recitation tempo and in the realisation of prolongation (madd) rules.

Table 3. Composition of the verse-level corpus by riwayat.

Riwayah	Reciters	Recordings	Duration (h)	Mean verse dur.\ (s)
Hafs 'an 'Asim	5	2,739	3.57	4.69
Warsh 'an Nafi'	5	2,562	3.78	5.30
Qalun 'an Nafi'	5	2,697	3.83	5.11
Total	15	7,998	11.17	5.03

Rule extraction from the canonical text identified 328 Idgham and 377 Qalqalah occurrences within Juz' 'Amma—705 unique textual occurrences before replication across reciters and riwayat (Table 4). Because each occurrence is realised by up to fifteen reciters, the number of acoustic instances available for learning is substantially larger than the textual counts, enabling the same phenomenon to be studied under diverse speakers and riwayat. Recording duration is balanced across riwayat (32–34% each), whereas subtype frequencies are strongly skewed: Lam Shamsiyyah accounts for more than 72% of Idgham occurrences while Idgham Lam-Ra occurs once, and Qalqalah Kubra (12 occurrences) is far rarer than Qalqalah Wusta (145).

Table 4. Unique textual occurrences of the annotated Tajweed subtypes in Juz' 'Ammā. Subtypes marked † constitute the five-class recognition task of Sect. 7; the remainder are annotated but excluded from the present classification experiments for the reasons stated in Sect. 7.

Idgham subtype	Occ.	Qalqalah subtype	Occ.
Lam Shamsiyyah	237	Qalqalah Wusta [†]	145
Idgham bi-Ghunnah [†]	65	Qalqalah Sughra [†]	123
Idgham bila-Ghunnah [†]	25	Qalqalah Mushaddadah [†]	97
Idgham Lam-Ra	1	Qalqalah Kubra	12
Total	328	Total	377

5 Annotation Methodology

Annotation provides rule-specific labels at the level of individual ayat. The protocol derives from established Tajweed references, whose linguistic and phonetic descriptions were translated into explicit, consistently applicable annotation criteria, and combines automatic candidate identification with manual acoustic verification and expert review.

5.1 Idgham

Idgham denotes the assimilation of one consonant into a following consonant, here primarily nun sakinah and tanwin before the letters designated for Idgham, optionally with nasalisation. Acoustically it is expected to remove or reduce an independent realisation of the first consonant, increase temporal overlap between adjacent segments, and yield a boundary dominated by the following consonant, with added low-frequency energy and nasal resonance when ghunnah is present. Candidates were extracted from the Qur'anic text by a Python script encoding the formal Tajweed conditions; the waveform and spectral representation of each corresponding recording were then inspected to localise the realisation and confirm the assimilation. Boundaries were placed to capture the acoustic event while excluding unrelated segments, with particular care where coarticulation, recording quality, rate or context made the boundary ambiguous.

5.2 Qalqalah

Qalqalah is the echo-like release produced by *qaf*, *ta'*, *ba'*, *jim*, *dal* in states requiring the rule, notably sukun states, without an independent vowel. Its signature is a distinct consonantal release: a transient energy increase, a clear burst, rapid spectral change and a short transition into the neighbouring segment. Candidates were identified textually from the relevant letters and contextual conditions, after which the temporal region of the release was located manually and verified as eligible.

Because the realisation varies with reciter, articulation, recording conditions and context, acoustic evidence was always interpreted jointly with the textual condition, and genuine Qalqalah was distinguished deliberately from ordinary consonantal release.

5.3 Workflow, Consistency and Validation

The workflow comprises four steps: *rule specification*, converting Tajweed definitions into annotation criteria; *automatic candidate extraction* from the textual material; *manual acoustic annotation* of temporal boundaries and acoustic parameters; and *expert verification* of rule applicability and segmentation. Consistency was enforced by deterministic textual rules for candidate generation and by predefined boundary criteria applied uniformly. All labels then underwent expert review, with uncertain boundaries, weak realisations and ambiguous conditions examined individually; an annotation was retained only when linguistic condition and acoustic realisation were judged compatible. Inter-annotator agreement was quantified on a stratified sample of 978 instances (15% of the 6,517 retained instances, proportional to subtype and riwayat) annotated independently by two trained annotators. Categorical subtype identification reached Cohen's $\kappa = 0.86$ (95% CI 0.83–0.89); adding the adjudicating expert as a third rater gave Fleiss' $\kappa = 0.84$. Temporal localisation was scored by boundary offset against the adjudicated reference: 91.4% of onsets and 89.7% of offsets lay within a ± 25 ms tolerance window (mean absolute offsets 14.2 ms and 17.8 ms); segments outside tolerance and disputed labels were re-examined jointly and resolved by the adjudicating expert. The ground truth thus combines rule-based extraction, acoustic localisation and expert judgement—essential because a rule's presence in the text does not determine its acoustic realisation.

6 Data Analysis

6.1 Acoustic Feature Representation

Each annotated segment is described by eight handcrafted descriptors chosen to reflect the phonetic mechanisms of the target rules: RMS mean (signal energy), zero-crossing-rate (ZCR) mean (temporal fine structure), spectral-centroid mean (spectral distribution), spectral-flux mean (spectral change), mean and peak nasality (nasal coupling), a depth metric (relative prominence and internal structure of the acoustic event), and duration (temporal extent of the rule-related event). Segment-level statistics are used so that every descriptor is defined uniformly across instances of differing length, and each corresponds to an acoustic dimension that Tajweed theory predicts to be affected by assimilation, nasalisation or consonantal release (Table 5).

Table 5. The eight acoustic descriptors and their relation to Tajweed realisation.

Descriptor	Acoustic dimension	Tajweed relevance
RMS mean	signal energy	overall intensity of the rule event
ZCR mean	temporal fine structure	noise-like transition at assimilation
Spectral-centroid mean	spectral distribution	brightness change at release
Spectral-flux mean	spectral change	rapid transition of Qalqalah release
Mean nasality	average nasal coupling	ghunnah of Idgham bi-Ghunnah
Peak nasality	maximal nasal coupling	strength of nasalisation
Depth metric	prominence / internal structure	articulatory depth of the event
Duration	temporal extent	assimilation and prolongation timing

6.2 Redundancy, Normalisation and Cleaning

Correlation analysis indicates partial redundancy: the correlation between ZCR and spectral centroid reaches 0.98, mean and peak nasality correlate at approximately 0.63, and spectral centroid and spectral flux show a moderate positive relationship. Correlated descriptors are not interchangeable, but redundancy motivates normalisation and subset evaluation. Features are standardised as

$$Z_j = \frac{x_j - \mu_j}{\sigma_j}, \quad (1)$$

where μ_j and σ_j are the mean and standard deviation of descriptor j over the training partition *only*; the same constants are then applied unchanged to the validation and test partitions, so no statistic of held-out data influences the transformation and no leakage path opens into distance- and margin-based models. Descriptors aggregate 25 ms frames (10 ms hop) at segment level. A cleaning stage then removed unreliable rule instances. A candidate instance was discarded if (i) its duration fell outside [120 ms, 12 s] or beyond $\pm 3\sigma$ of its subtype's log-duration mean, (ii) its forced-alignment confidence was below 0.80, (iii) its segmental signal-to-noise ratio, estimated by spectral subtraction, was below 15 dB, or (iv) clipping or dropouts affected more than 1% of its samples. These checks removed 308 of the 6,825 candidate instances (4.5%), leaving 6,517 instances for modelling; the reduction fell on all five classes and most visibly on the majority classes, while Idgham bila-Ghunnah remained substantially under-represented.

Class distribution. The five-class task comprises the 6,517 retained instances of the five marked subtypes: Idgham bi-Ghunnah 931, Idgham bila-Ghunnah 352, Qalqalah Mushaddadah 1,392, Qalqalah Sughra 1,761 and Qalqalah Wusta 2,081, i.e. a 5.9:1 ratio between the largest and smallest class. At corpus level the riwayat balance

limits riwayat-specific bias, but the subtype distribution is markedly imbalanced: models optimised for overall accuracy may favour dominant classes while failing on rare but linguistically important phenomena, so evaluation must rely on class-sensitive metrics alongside accuracy (Sect. 7.4). The multi-reciter, multi-riwayah design additionally permits testing whether learned representations capture Tajweed-related rather than reciter- or riwayat-specific patterns.

7 Machine Learning Framework

7.1 Task Formulation

Recognition is formulated as supervised multiclass classification: each segmented instance is assigned to one of five Tajweed classes—Idgham bi-Ghunnah, Idgham bila-Ghunnah, Qalqalah Mushaddadah, Qalqalah Sughra and Qalqalah Wusta—from the descriptors of Sect. 6. These five classes are the subtypes of Table 4 retained for the classification experiments. The three annotated but unmodelled subtypes are excluded on two grounds. Idgham Lam-Ra (1 textual occurrence) and Qalqalah Kubra (12 occurrences) yield at most 15 and 180 acoustic instances respectively; at such support, per-class recall estimates carry confidence intervals wider than the effects this study resolves, and any statistic for these classes would be dominated by a handful of reciters. Lam Shamsiyyah is, conversely, over-represented (237 occurrences, 72% of Idgham): it assimilates the lam before sun letters, a different articulatory site from the nun-sakinah and tanwin assimilations modelled here, and including it would let one dominant class mask minority behaviour in aggregate metrics. Both decisions are reported in Table 4 and revisited in Sect. 11.

7.2 Classifiers

Five complementary algorithms cover distinct learning paradigms. The soft-margin support vector machine (SVM) minimises $\frac{1}{2}\|\mathbf{w}\|^2 + c\sum \xi_i$ subject to $\mathbf{y}_i(\mathbf{w}^T\phi(\mathbf{x}_i) + b) \geq 1 - \xi_i$, combining binary decision functions for the five-class task; it suits compact descriptors but is sensitive to scaling and kernel choice. The random forest (RF) aggregates bootstrap-trained trees over randomised feature subsets,

$$\hat{\mathbf{y}} = \text{mode}\{T_b(\mathbf{x})\}_{b=1}^B, \quad (2)$$

modelling nonlinear descriptor interactions and providing importance estimates. k -nearest neighbours (KNN) assigns the majority class among the k closest training observations,

$$\hat{\mathbf{y}} = \text{mode}\{y_i : \mathbf{x}_i \in \mathcal{N}_k(\mathbf{x})\}, \quad (3)$$

serving as an instance-based baseline sensitive to local acoustic similarity. XGBoost [21] builds an additive ensemble $\hat{\mathbf{y}}_i = \sum f_t(\mathbf{x}_i)$ over regression trees, optimising a regularised objective $\mathcal{L} = \sum l(y_i, \hat{\mathbf{y}}_i) + \sum \Omega(f_t)$, which captures higher-order interactions among temporal, spectral and phonatory descriptors while remaining effective for

small tabular feature sets. Finally, a multilayer perceptron (MLP) applies hidden-layer transformations $\mathbf{h} = \sigma(\mathbf{W}\mathbf{x} + \mathbf{b})$ with a softmax output $P(y = k|\mathbf{x}) = \exp(z_k) / \sum_i \exp(z_i)$, testing whether nonlinear combinations of the descriptors add discriminative information. Together these models evaluate whether the acoustic representation remains effective across substantially different learning paradigms.

7.3 Class-Imbalance Handling

Imbalance was investigated through five configurations: unbalanced training, class weighting, SMOTE [22], Borderline-SMOTE, and SMOTE with class weighting. SMOTE synthesises minority observations by interpolating between neighbouring minority samples,

$$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda(\mathbf{x}_{\text{nn}} - \mathbf{x}_i), \lambda \in [0, 1], \quad (4)$$

whereas Borderline-SMOTE restricts synthesis to samples near class boundaries, using $k = 5$ defining neighbours and synthetic ratios of 50% and 100% of each minority class's training count. All configurations are evaluated with conventional and imbalance-sensitive metrics, since high overall accuracy can coexist with poor minority-class recognition.

7.4 Evaluation Metrics

Performance is measured by accuracy, per-class precision, recall and F1-score, Macro-F1, weighted F1, balanced accuracy and one-vs-rest ROC-AUC. With $K = 5$ classes, Macro-F1 is the unweighted mean of per-class F1-scores and balanced accuracy the mean of per-class recalls; precision-recall (PR) analysis complements ROC analysis because ROC-AUC can remain high for rare classes even when precision deteriorates at practical operating points.

8 Experiments

8.1 Experimental Protocol

Seven experiment families address the research questions: **E1** evaluates the five classifiers on the cleaned dataset with the proposed representation; **E2** compares it against an MFCC + delta + delta-delta baseline under identical protocols; **E3** analyses per-class behaviour through normalised confusion matrices; **E4** examines discriminability with one-vs-rest ROC and PR curves; **E5** quantifies descriptor contributions through tree-based importance and ablation; **E6** compares the imbalance-handling configurations of Sect. 7.3; and **E7** tests cross-riwayah generalisation by training on one riwayah and evaluating on another for all six ordered pairs. Uncertainty is estimated by bootstrap resampling of the test set (1,000 resamples). Table 6 summarises the design.

Table 6. Experimental design: question, protocol and primary metrics of each experiment family.

ID	Question	Protocol	Primary metrics
E1	Does the representation support recognition?	5 classifiers, cleaned data	Acc., bal. acc., Macro-F1
E2	Is it better than MFCC?	identical protocol, MFCC baseline	Macro-F1, Acc.
E3	Which classes drive performance?	normalised confusion matrices	per-class recall
E4	How separable are the classes?	one-vs-rest ROC / PR	ROC-AUC, PR
E5	Which descriptors matter?	importance + ablation	Macro-F1 change
E6	Does balancing help?	5 training configurations	bal. acc., minority recall
E7	Does it transfer across riwayat?	6 train→test pairs	Macro-F1

Protocol and reproducibility. All experiments use a reciter-disjoint stratified split: the fifteen reciters are partitioned into nine training, three validation and three test reciters (60/20/20 at instance level), stratification is enforced over subtype $\times$ riwayat cells so that every partition preserves the class and riwayat mixtures, and no reciter appears in two partitions, which removes speaker- and recording-level leakage by construction. The test partition is held out until final evaluation and model selection uses the validation partition. The unbalanced-versus-SMOTE comparison additionally uses grouped stratified five-fold cross-validation with reciters as groups (three reciters per fold), so each fold remains speaker-disjoint. Splits, SMOTE draws and model initialisations derive from seed 42, and every configuration repeats the identical protocol. Hyperparameters are selected on the validation partition by Macro-F1 from the grids: SVM—RBF and linear kernels, $C \in \{0.1, 1, 10, 100\}$, $\gamma \in \{0.01, 0.05, 0.1, 0.5\}$; random forest—200/500/1,000 trees at depth 8/12/16/unrestricted; KNN— $k \in \{3, 5, 7, 9, 11\}$; XGBoost—200/400/600 estimators at learning rate 0.05/0.1/0.2 and depth 3/5/7; MLP—hidden layers (32,16)/(64,32)/(128,64,32) with dropout 0.2/0.3/0.4 and learning rate 1e-3/3e-4, early stopping with patience 15. The selected configurations are SVM (RBF, $C = 10$, $\gamma = 0.05$), random forest (500 trees, depth 12), KNN ($k = 5$), XGBoost (400 estimators, learning rate 0.1, depth 5) and MLP ((64,32), dropout 0.3, learning rate 1e-3). The implementation uses Python 3.11 with scikit-learn 1.4.2, xgboost 2.0.3, librosa 0.10.1 and NumPy 1.26.4 on a single CPU node (2 $\times$ Xeon Silver 4214, 48 GB RAM); the full suite, including all balancing configurations and cross-riwayat pairs, runs in under 40 minutes without a GPU.

9 Results

9.1 Overall Classifier Performance

Table 7 reports the primary comparison (class-weighted configuration). XGBoost achieves the highest Macro-F1 (0.6935) with accuracy 0.7202 and balanced accuracy 0.6953; SVM attains the highest balanced accuracy (0.7126) and random forest the highest accuracy (0.7298), while MLP is weakest on Macro-F1 (0.6219). The divergence between accuracy and Macro-F1—random forest leads on accuracy yet trails XGBoost by 0.0252 on Macro-F1—confirms that accuracy alone mischaracterises this imbalanced problem.

Table 7. Classifier comparison on the held-out test set with the proposed representation (class-weighted configuration). Macro-recall equals balanced accuracy by definition and is reported once.

Model	Accuracy	Balanced acc.	Macro-prec.	Macro-F1	Weighted F1
SVM	0.7083	0.7126	0.6902	0.6920	0.7080
Random forest	0.7298	0.6668	0.7036	0.6683	0.7192
KNN	0.7179	0.6632	0.6872	0.6687	0.7125
XGBoost	0.7202	0.6953	0.6951	0.6935	0.7195
MLP	0.7071	0.6356	0.7001	0.6219	0.6873

9.2 Proposed Representation Versus MFCC Baseline

The proposed descriptors improve Macro-F1 for every classifier (Fig. 2), with the largest gains for random forest (+0.1348), KNN (+0.1336) and XGBoost (+0.1191); accuracy rises from 0.5869 to 0.7298 (random forest), 0.5774 to 0.7179 (KNN) and 0.6119 to 0.7202 (XGBoost). Consistency across all five paradigms indicates that the advantage originates in the information content of the representation rather than in an interaction with a particular decision boundary.

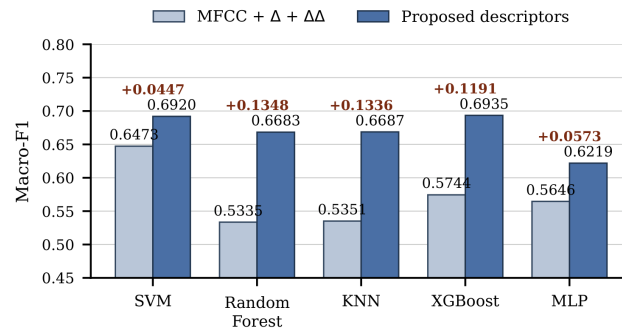


Fig. 2. Macro-F1 of the proposed acoustic descriptors and of the MFCC + delta + delta-delta baseline for each classifier; annotations give the absolute gain of the proposed representation.

9.3 Per-Class Behaviour, Confusion and Discriminability

Per-class results differ sharply (Fig. 3). Idgham bi-Ghunnah is recognised almost perfectly by every classifier (recall ≈ 1.00 , ROC-AUC 1.00), and Qalqalah Wusta remains strong and architecturally stable across all five models (recall 0.88–0.92, ROC-AUC 0.99). Qalqalah Mushaddadah is intermediate (recall 0.58–0.72, ROC-AUC 0.89) and Qalqalah Sughra is recovered reliably by the strongest model (recall 0.55, ROC-AUC 0.84). Idgham bila-Ghunnah is the most difficult class, with recall spanning 0.017–0.49 across models and ROC-AUC 0.87 for XGBoost; its errors are distributed across several competitors rather than concentrated in one, indicating overlap with multiple categories under the current representation. SVM’s relatively high recall for Idgham bila-Ghunnah (0.49) explains its leading balanced accuracy despite XGBoost’s higher Macro-F1. PR analysis shows precision for Idgham bila-Ghunnah falling steeply as recall increases, so its ROC-AUC must not be read as uniform strength: the class is ranked above chance, yet only part of it can be recovered without admitting many false positives. With 352 retained instances this class also carries the widest bootstrap uncertainty of the five, so the ranges of Fig. 3 are evidence of systematic difficulty rather than precise per-model rankings.

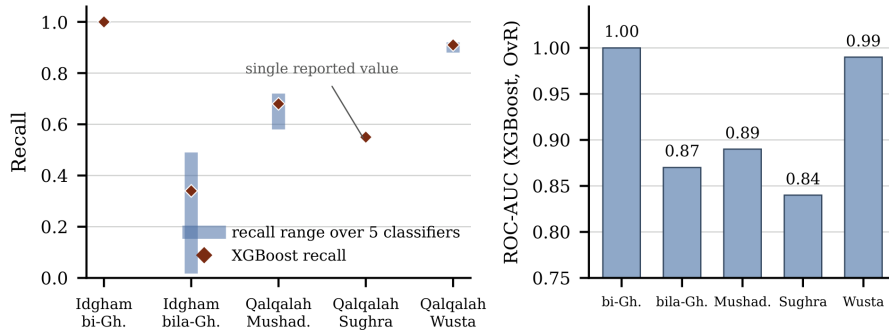


Fig. 3. Per-class discriminability of the five Tajweed categories, with the Idgham subtypes followed by the Qalqalah subtypes. Left: recall attained by the five classifiers, shown as the range across models with the XGBoost value marked; Qalqalah Sughra carries a single reported value. Right: ROC-AUC of XGBoost for the same classes.

9.4 Feature Importance and Ablation

In XGBoost, duration dominates feature importance (≈ 0.52) followed by the depth metric (≈ 0.21); in random forest the same two descriptors lead (depth ≈ 0.33 , duration ≈ 0.24) over a broader distribution. Ablation confirms their causal contribution (Fig. 4): removing the depth metric reduces Macro-F1 from 0.6935 to 0.5169 (-0.1766) and removing duration to 0.5587 (-0.1348), whereas retaining duration and depth alone still yields 0.6424 and nasality alone collapses to 0.2290,

i.e. no better than random over five classes. Removing the spectral descriptors (centroid and flux) costs only 0.0423 Macro-F1, and removing the strongly correlated nasality pair 0.0391, so both groups carry information that complements—but does not duplicate—the temporal descriptors. The full set remains superior to every subset and every removal variant: the descriptors are not interchangeable, a small subset carries most of the discriminative signal, and the residual classes contribute the difference between a good and the best result.

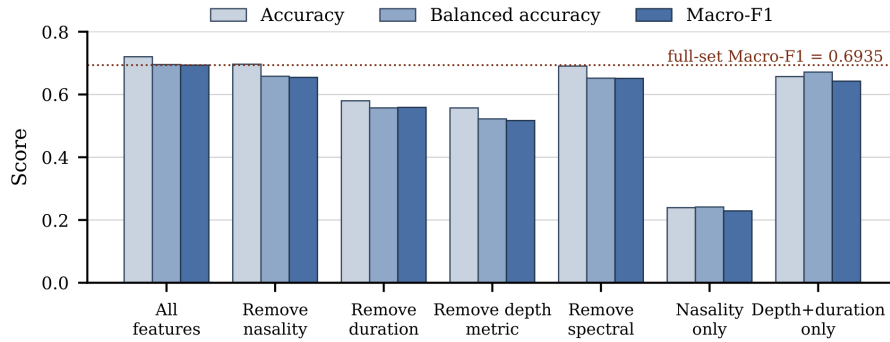


Fig. 4. Feature ablation with XGBoost on the held-out test set. Macro-F1, balanced accuracy and accuracy for seven configurations: the full descriptor set, four removal variants and two restricted subsets. Protocol, seed and class weighting are identical throughout.

Class-balancing results. Balancing trades overall accuracy for minority-class sensitivity (Fig. 5). Unbalanced training yields the highest accuracy (0.7357) but an Idgham bila-Ghunnah recall of only 0.0678 (F1 0.1111). Class weighting raises that recall to 0.3390 and Macro-F1 to 0.6935 at a cost of 1.55 points of accuracy, and SMOTE gives the strongest balanced performance: balanced accuracy 0.6617 $\rightarrow$ 0.7121, Macro-F1 0.6592 $\rightarrow$ 0.6951, and Idgham bila-Ghunnah recall 0.0678 $\rightarrow$ 0.4915 (+42.4 points). Borderline-SMOTE is less effective than standard SMOTE, and adding class weighting after SMOTE yields no further change, indicating that boundary-focused synthesis does not suit this feature space.

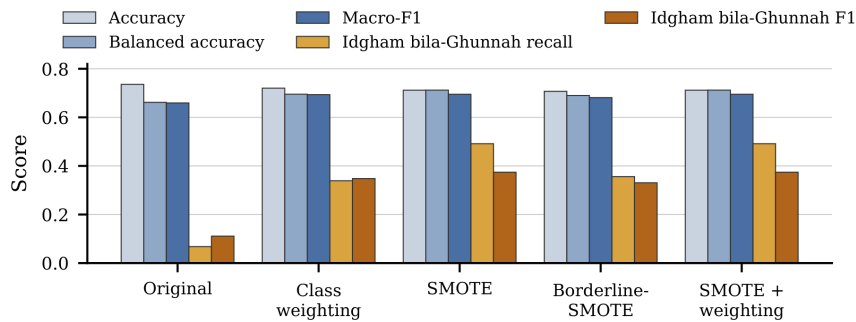


Fig. 5. Effect of class-balancing strategies with XGBoost. Accuracy, balanced accuracy, Macro-F1 and the recall and F1 of the rarest class, Idgham bila-Ghunnah, for the original training set, class weighting, SMOTE, borderline-SMOTE and SMOTE combined with class weighting.

9.5 Cross-Riwayah Generalisation and Uncertainty

Training on one riwayah and testing on another yields Macro-F1 of 0.6584–0.6718 across the six transfer directions (Fig. 6): a narrow 0.0134 range, best for Qalun $\rightarrow$ Warsh (0.6718) and weakest for Qalun $\rightarrow$ Hafs (0.6584), consistently below the within-distribution result, i.e. partially riwayah-independent information alongside residual distribution shift. Bootstrap resampling gives a mean Macro-F1 of 0.6934 with a 95% confidence interval of [0.6604, 0.7240]. McNemar and Wilcoxon signed-rank tests are reserved for a multi-seed replication study: with the single seed used here the bootstrap intervals of the leading models overlap (Sect. 9), so no pairwise superiority claim is made.

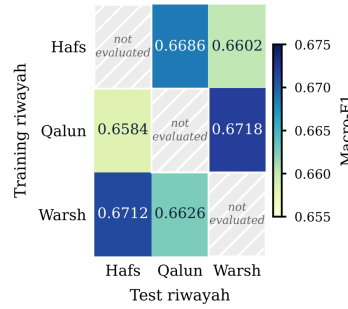


Fig. 6. Cross-riwayah generalisation (Macro-F1): rows are training riwayat and columns testing riwayat. Diagonal cells (within-riwayah evaluation) correspond to the pooled results of Table 7 and are not repeated here.

10 Discussion

10.1 Why Interpretable Descriptors Outperform MFCC Here

The consistent gain of the proposed descriptors over the MFCC baseline, across five learners of very different inductive bias, localises the improvement in the representation rather than in the models. The reason is structural: cepstral coefficients summarise the short-term spectral envelope, and frame-level averaging smooths precisely the transient release bursts and boundary reorganisations that define Qalqalah and Idgham, while nasal coupling enters them only indirectly. The proposed set makes those dimensions explicit—temporal extent, release dynamics and nasal coupling—so even an instance-based learner separates classes that a cepstral summary conflates. Two guards against overreading this result: only one MFCC configuration (with deltas) was evaluated, and the claim is task-specific, not a

general superiority of handcrafted descriptors over learned representations; on larger corpora, self-supervised embeddings [10, 11] remain the natural next comparison, and the present evidence predicts that they would need to recover the same rule-relevant dimensions to succeed here.

10.2 Duration and Depth as Acoustic Evidence of Tajweed

Duration and the depth metric converge across two independent forms of evidence—tree-based importance and controlled removal—which makes an incidental explanation unlikely. The convergence is phonetically coherent: assimilation redistributes energy across a boundary and therefore changes the temporal footprint of the event, while Qalqalah inserts a release that restructures the event's internal prominence. Their dominance supports the paper's working premise that Tajweed rules are acoustically observable phonetic instructions rather than purely symbolic text properties. Because duration is also the descriptor most sensitive to boundary placement (Sect. 11), its importance must be read jointly with the boundary-tolerance analysis of Sect. 5, which bounds that sensitivity at a few tens of milliseconds.

10.3 Why Idgham bila-Ghunnah Remains Difficult

Idgham bi-Ghunnah carries an explicit nasal marker that the nasality descriptors capture directly; Idgham bila-Ghunnah lacks it and must be recognised from subtler temporal-spectral assimilation cues that overlap both neighbouring phonetic events and other target classes. The SMOTE result decomposes the difficulty: part is statistical, since synthetic coverage of the feature space raises recall from 0.0678 to 0.4915, and the residual confusion after balancing is therefore acoustic overlap that no classifier change within this representation will remove. The steep precision-recall trade-off of this class is the practical signature of that overlap and should guide deployment: assistive systems can trade recall for precision explicitly at this class.

10.4 Cross-Riwayah Transfer as Managed Domain Shift

The cross-riwayah drop of two to three Macro-F1 points is far below the between-class margins the models exploit. Two conclusions follow: the descriptors encode rule-related information that is partly tradition-invariant, which is necessary for any practical assessment tool; and the residual, consistent degradation identifies riwayah as a measurable domain shift in articulation, tempo and prosody, most visible in duration-sensitive descriptors. Riwayah should therefore enter system design explicitly—as a conditioning variable, an adaptation axis or an invariance constraint—rather than being pooled and hoped away.

10.5 Educational Implications

For computer-assisted recitation training, rule-level localisation changes the feedback unit from a global score to an event: a suspected rule, its temporal position and the acoustic dimension at fault (nasalisation, release, timing). Because the system distinguishes riwayat-legitimate variation from error only within the three riwayat studied, feedback must be presented as assistive evidence with a confidence indication, and borderline bila-Ghunnah decisions should be routed to expert review rather than shown as verdicts.

10.6 Scientific Implications and Positioning

Methodologically, the study supports modelling Tajweed recognition as a structured, rule-aware problem: categories differ in articulatory mechanism and acoustic salience, so representations preserving rule-specific dimensions outperform undifferentiated ones, and domain knowledge remains valuable even with conventional learners. The results also fit a broader pattern in small-data acoustics, where interpretable tabular descriptors with gradient-boosted trees remain competitive with deeper architectures: the contribution here is evidence that rule-level annotation plus targeted descriptors makes a linguistically fine-grained acoustic class structure learnable at all.

11 Limitations

Corpus scope. The corpus covers Juz' 'Amma and two rule families; observed descriptor behaviour may not transfer to other surahs or rules, and fifteen reciters across three riwayat cannot establish corpus-independent generalisation. All gains are therefore claims within the experimental domain, and the corpus cards report coverage explicitly so that reuse decisions can be made informedly. **Annotation uncertainty.** Assimilation produces gradual acoustic transitions, so boundaries are inherently less discrete than for plosives; because duration and depth depend on boundary placement, small shifts perturb those descriptors for acoustically equivalent recitations. The tolerance-window analysis (Sect. 5) quantifies this uncertainty but does not eliminate it, and probabilistic boundaries remain the appropriate long-term treatment. **Class imbalance and rare subtypes.** Natural subtype frequencies are skewed, and two annotated subtypes were excluded for insufficient support (Sect. 7); conclusions about those subtypes, and about per-class ranks within the smallest modelled class, are correspondingly limited. **Synthetic oversampling.** SMOTE interpolates in feature space rather than adding authentic recitations, so part of its gain reflects better feature-space coverage, not new phonetic diversity; class-aware acquisition of genuine minority examples remains necessary. **Speaker variability and leakage.** The reciter-disjoint protocol removes speaker leakage by construction within this corpus, but fifteen professional male reciters do not span the variability of real learners—gender, age, proficiency, pathology—which deployed systems will encounter. **Domain shift.** Recording conditions, microphones and compression differ across sources, and the cross-riwayat experiment shows that

even legitimate transmission variation shifts feature distributions; robustness to unseen corpora is untested. **External and statistical validity.** Results derive from one corpus and one seed; bootstrap intervals quantify test-set uncertainty, but pairwise classifier comparisons remain provisional pending multi-seed replication with McNemar and Wilcoxon tests. **Representation scope.** Segment-level statistics cannot encode long-range dependencies or transitions between neighbouring phonemes, exactly where some assimilation evidence may reside; sequential or attention-based architectures, and self-supervised embeddings, should be evaluated alongside—not simply instead of—the interpretable descriptors.

Future directions. Three follow-ups address these limits directly. First, evaluating the identical protocol on the Hafs-only subset of an independent public recitation corpus would separate corpus-specific from representation-specific behaviour. Second, a riwayat-conditioned model—with the reciter's riwayat supplied as an explicit input or with per-riwayat fine-tuning—should reduce the cross-riwayat penalty without enlarging the balanced subset. Third, phoneme-aligned rather than fixed-window segmentation would remove the boundary noise that currently caps the recall of Qalqalah Mushaddadah and would make the duration descriptor directly interpretable in terms of Tajweed timing.

12 Conclusion

This paper formulated automatic Tajweed recognition as a riwayat-aware, fine-grained acoustic classification problem and instantiated it end to end: a 7,998-recording, 11.17 h corpus of Juz' 'Amma spanning Hafs, Warsh and Qalun; expert-verified rule-level annotation of Idgham and Qalqalah; an interpretable eight-descriptor representation; and a five-class evaluation across five learning paradigms. The representation outperformed an MFCC baseline for every classifier (best Macro-F1 0.6935, XGBoost), ablation established duration and the depth metric as dominant cues, balancing materially improved the difficult Idgham bila-Ghunnah class, and cross-riwayat transfer remained within 0.6584–0.6718. Future work will expand the corpus in surahs, reciters, riwayat and rules; quantify annotation agreement; adopt speaker- and corpus-independent evaluation; and investigate sequential, attention-based and multimodal models combining acoustic evidence with textual and phonological constraints, towards interpretable, confidence-aware recitation support that complements expert instruction.

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